Source Use and Cultural Image Questions: Small External-Label Pilots in Yoruba and Swahili

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Multilingual evaluation can confound question comprehension, answer language, supplied evidence, and cultural knowledge. We report two small pilots that separate some of these factors using published human labels. In AfriQA, 24 questions, equally divided between Yoruba and Swahili, are each evaluated with a native question and English passage, the published English question and the same passage, and a native question with no passage. Every condition explicitly requires an English source span or JSON null. In a separate Afri-MCQA study, 12 Yoruba image groups are tested with native and English question-option pairs, plus a native-language condition without the image. Two local model configurations complete 234 requests, including 18 interface checks excluded from locked results. Native-question source exact matches are 3/12 Yoruba and 7/12 Swahili for Qwen3.5, and 7/12 and 6/12 for Gemma4; English-question differences are mixed. Both models obey all 24 empty-passage controls. Cultural-image accuracy is 5/12 and 7/12, compared with 2/12 and 3/12 without images. These descriptive results support a reproducible evaluation interface and show why schema compliance, extraction accuracy, and cultural image use should be reported separately. They do not establish native-language service quality or new community validation.

Additional Key Words and Phrases: multilingual evaluation, Yoruba, Swahili, source grounding, cultural question answering, abstention

1 Introduction and Background

A multilingual assistant must understand a question, use suitable evidence, and produce an answer appropriate to its user. A score that combines these steps can conceal which component failed. Even a controlled comparison becomes ambiguous if a native-language question is graded against an English reference without telling the model which answer language to use. Likewise, an image question can reward knowledge or option-based guessing when the image itself contributes little. Small experiments should make these contracts and controls explicit before supporting a larger service claim.

Published resources supply stronger starting points than newly generated translations with assumed correctness. AfriQA links African-language information-seeking questions to cross-lingual answer content [5]. IrokoBench provides human-translated reasoning and knowledge tasks across African languages [1]. Afri-MCQA adds culturally situated images and native-speaker-created question-answer pairs [12]. These resources answer different questions: translated knowledge tests, supplied-source extraction, and culturally situated visual questions should not be pooled into one language score.

Human assessment also remains necessary when automatic correctness is insufficient. MEMERAG studies multilingual retrieval-augmented generation through expert judgments of relevance and faithfulness [3]. Howard's cultural-routing and retrieval-fusion systems provide implementation experience for a future local-evidence assistant [2, 11]. The present study does not reproduce their full systems. It establishes an auditable baseline using existing labels, paired inputs, and simple deterministic scoring.

We ask two bounded questions. With a fixed English source and output contract, does changing the question language change extraction accuracy? With a fixed cultural item and option order, how do the language of questions and options, and image availability, change multiple-choice accuracy? The contribution is the frozen pilot design and artifact, including mixed results and a clear account of what remains untested. Code, recorded outcomes, source identifiers, and the manuscript package are available at https://github.com/Saurav-K-Aryal/multilingual-grounding-pilot.

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2 Methods

2.1 Sources, selection, and evaluation units

The protocol was frozen before the new model calls and approved independently of those outcomes. It contains 36 locked units: 24 AfriQA questions and 12 Afri-MCQA image groups. Three additional units serve only as interface checks. Each unit has three conditions and is evaluated on two models, giving $36 \times 3 \times 2 = 216$ locked responses and $3 \times 3 \times 2 = 18$ interface responses. The 234 requests do not represent 234 independent questions.

For AfriQA we use the official Yoruba and Swahili gold-passage test files from Git revision `5c2993375369` (full digest in the artifact). Eligibility requires a matching official question record marked `human_translation`, nonempty native and English questions, a title, a 40–2,200-character passage, and at least one official English answer span verified at its recorded character offset. Eligible spans are at most 100 characters and 12 whitespace-delimited words. This excludes answer types that cannot be evaluated by the declared extractive contract; it does not define the full AfriQA task.

Using seed 20261031, we sample three questions in each of four English-question-prefix strata per language: *who*, *when*, *where*, and *other*. One further question per language is reserved for interface checks. English passage titles are unique across all 26 selected questions, including the interface and locked partitions. The exclusions and sampling order are preserved. This is a small stratified subset, not an estimate of benchmark-wide performance. Yoruba and Swahili have different source questions, so their raw accuracies are not a causal language comparison. Native-versus-English comparisons use the same item.

For Afri-MCQA, the public Yoruba test records expose options without gold labels. We therefore use gold-bearing official development records and create an internal research holdout. We do not report official test-set accuracy. The actual dataset-server revision is `d5d4db17fc27` (full digest in the artifact), which differs from the latest repository revision observed during collection. The two are recorded separately.

We select three distinct image groups from each of four metadata categories: cooking and food; objects, materials, and clothing; geography, buildings, and landmarks; and brands, products, and companies. A thirteenth group is reserved for interface checks. Empty or duplicate options and person-identification questions outside the object-centric scope are excluded. Source image-group identifiers and decoded-image hashes are disjoint between interface and locked partitions. The source snapshot contains Yoruba but not Swahili, so the cultural experiment is Yoruba-only. Its native-speaker authorship belongs to the external resource, not to new annotation performed in this project.

2.2 Paired conditions and response contracts

AfriQA has three conditions: a native question with its gold English passage; the published English question with the identical passage; and the native question with an empty passage. English system instructions explicitly require a JSON object with the single key `answer`. A non-null answer must be an English span copied from the supplied passage, regardless of question language. An empty passage requires JSON null even when the model knows the answer. This isolates cross-lingual question comprehension and English-source extraction; it does not measure the quality of a natural Yoruba or Swahili answer.

Afri-MCQA has an image/native-question condition, an image/English-question condition, and a no-image/native-question condition. We retain the published correct and incorrect option pairs and apply one deterministic joint permutation to both languages. Correct positions are balanced at three each for A, B, C, and D over the 12 locked items. The two native conditions use identical question and option text. Images undergo EXIF orientation correction,

conversion to RGB, resizing to at most 768 pixels per edge, and JPEG encoding at quality 90. The same processed bytes are used in both image conditions and both models.

The multiple-choice response is a one-key JSON object containing an A–D letter. In the no-image condition the model is instructed to make its best choice. This is a knowledge/guessing control, not a demand to abstain on every missing-image question. English meta-instructions are shared across conditions; only the specified question, option language, or image changes.

2.3 Executed models and measurements

The executed server tags are `qwen3.5:4b` and `gemma4:26b`. The former uses an NVFP4 artifact with an MLX runner; the latter uses Q4_K_M with a llama.cpp runner and reports 25.2 billion total parameters. Upstream model cards describe the respective families [4, 10], while the artifact records the actual local weight digests and server metadata. Family, scale, quantization, and backend change together; the comparison does not isolate model age or architecture.

Models run sequentially through the native local chat endpoint with JSON formatting, temperature zero, seed 20261031, a 6,144-token requested context, a 128-token output cap, and thinking disabled. Each executes the same 117 frozen cases once. Interface cases occur first; locked case order is randomized independently of outcomes. There are no retries, replacement items, output repair, or model-based prompt selection. Raw responses and errors are preserved, and both models are explicitly unloaded afterward. This is not a memory, latency, or energy benchmark.

For supported AfriQA cases, primary exact match compares the trimmed returned string with a verified official gold span. Secondary normalized exact match lowercases text, removes ASCII punctuation and English articles, and normalizes whitespace. Token F1 is twice the multiset token overlap divided by the total predicted and gold token counts, maximized over eligible references. We also record whether a non-null answer occurs contiguously in the supplied passage. Empty-passage success requires valid JSON with a literal null. These are separate measurements: a source span may answer the wrong question, and a valid schema may contain an incorrect answer.

Afri-MCQA accuracy uses the frozen correct letter. Paired discordances count which individual items change correctness across conditions. We report descriptive counts and means without treating repeated conditions or models as independent observations. A separate CPU readback checks all 234 raw responses, prompt strings, 52 image payloads, official source texts and option mappings, and every reported metric. Source and public-artifact hashes identify any omitted input content or path-only normalization.

3 Results

3.1 Source extraction and empty-evidence controls

All 234 requests completed with valid one-key response schemas and no transport errors. The 18 interface responses remain excluded from the locked metrics. Table 1 reports exact-match counts and mean token F1 on the 12 locked questions per language and condition. Normalized exact-match counts equal strict span-match counts in these observations.

Question-language effects are mixed (Figure 1). For Qwen3.5's Yoruba items, one is correct only with the native question and three only with the English question; the net difference is two items favoring English. Its Swahili comparison has two native-only and one English-only success. Gemma4 has two native-only and one English-only success for Yoruba, and no correctness discordance for Swahili. These observations do not support a universal native-language deficit or a stable language ranking.

Table 1. AfriQA results on the same 12 questions within each row. Both question conditions require English source-span answers. F1 is a mean over the 12 questions, including zero for null answers.

Model tag	Question source	Native EM	English EM	Native F1	English F1
Qwen3.5	Yoruba	3/12	5/12	.404	.639
Qwen3.5	Swahili	7/12	6/12	.672	.588
Gemma4	Yoruba	7/12	6/12	.715	.632
Gemma4	Swahili	6/12	6/12	.557	.557

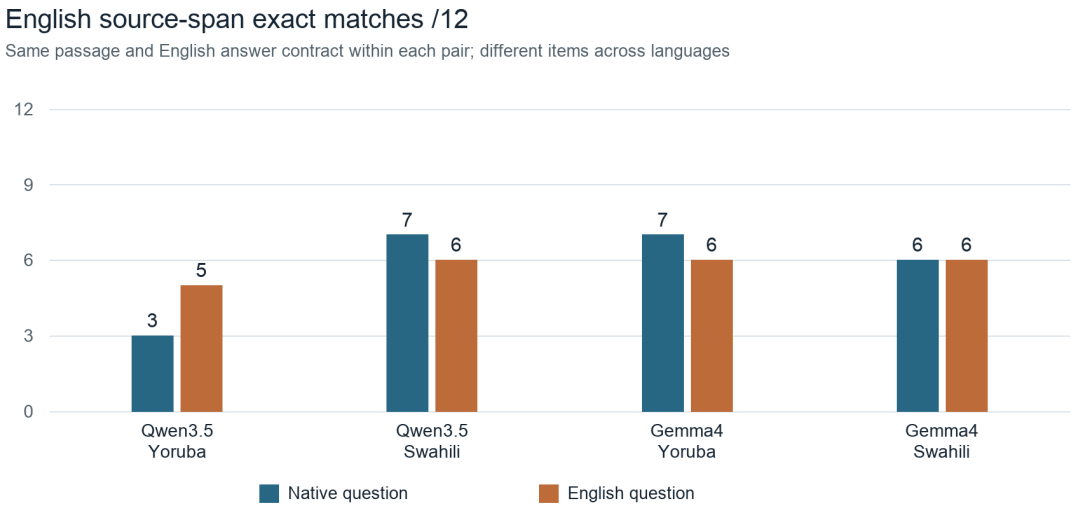


Fig. 1. Strict English-source exact matches. Every pair shares its question meaning, passage, and answer contract; Yoruba and Swahili use different items. Bars describe this selected subset rather than population uncertainty.

Both models return null in all 24 locked empty-passage cases, 12 per language. This verifies compliance with an explicit missing-input instruction. It does not establish calibrated uncertainty when a plausible passage is stale, contradictory, or from the wrong locality. Across the 48 supported conditions per model, Qwen3.5 answers 44 times and Gemma4 41 times. Of these answers, 43 and 41, respectively, occur as contiguous source spans. High span adherence therefore coexists with substantially lower reference-answer accuracy. We retain the single Qwen3.5 non-span answer rather than correcting it.

3.2 Cultural image use and language

The Yoruba cultural experiment gives a clearer image/no-image difference on the selected groups (Table 2 and Figure 2). Qwen3.5 answers 5/12 native image questions correctly, compared with 2/12 without images; Gemma4 answers 7/12 compared with 3/12. The paired differences comprise four gains and one loss for Qwen3.5, and five gains and one loss for Gemma4. Thus an image helps the aggregate count while making some individual decisions worse.

Both configurations reach 8/12 with the published English questions and options. Relative to native questions and options, Qwen3.5 has three English-only successes and no native-only success; Gemma4 has two English-only and

Table 2. Afri-MCQA: the same 12 Yoruba image groups, fixed option permutations, and public development labels. These are internal research-holdout results, not official hidden-label test scores.

Model tag	Image/native	Image/English	No image/native	Image net gain
Qwen3.5	5/12	8/12	2/12	+3
Gemma4	7/12	8/12	3/12	+4

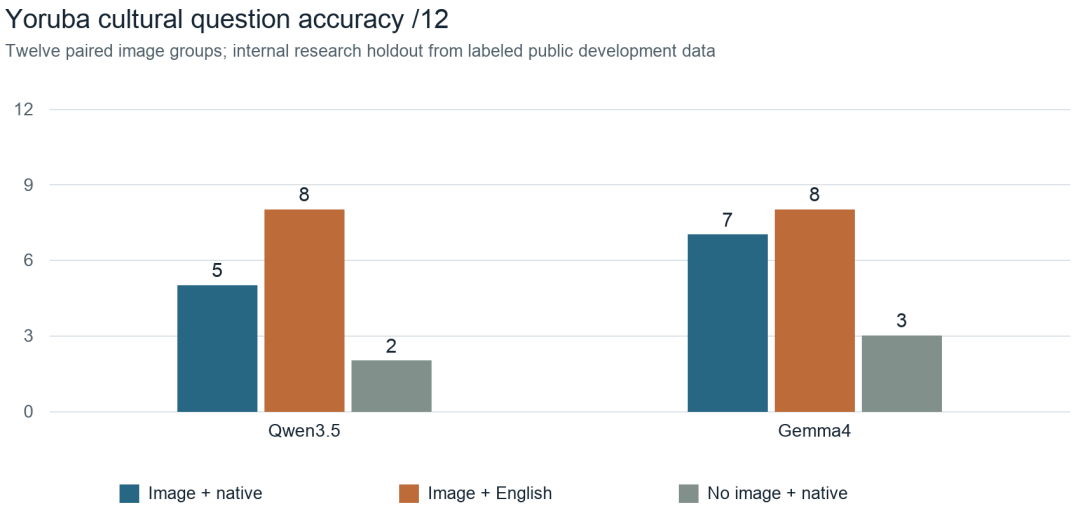


Fig. 2. Cultural question outcomes. The no-image control keeps native wording and options; the English-image condition uses published English questions and options. All three conditions share the same 12 image-group units.

one native-only success. The exact sample is too small to infer general cultural competence or the effect of language resources. Native-speaker creation of the original resource does not make these model outputs newly human-validated responses.

3.3 Validity and practical interpretation

The two studies measure different outputs and should remain separate. AfriQA grades English extraction under cross-lingual questions. Afri-MCQA grades a forced choice among published cultural options. Neither establishes natural native-language generation, appropriateness for a particular community, evidence freshness, source-locality selection, or an actual retrieval system. Empty evidence is a deliberately simple authored control. An unrelated but fluent passage is a more difficult future control.

The subset filters favor short extractive references, four question-prefix strata, and four cultural categories. A strict gold-span score may mark a semantically plausible alternative wrong; token overlap offers a limited secondary diagnostic, not a human adequacy judgment. Public benchmark contamination in pretrained models is not excluded. The cultural holdout comes from published development data because test labels were unavailable. There is one execution per configuration, no native-language reviewer recruited for this pilot, and no sampling-based claim about a broader

population. Backend and quantization differences also prevent attributing model differences to parameter scale or recency alone.

The practical result is measurement readiness: source compliance, correct extraction, question-language effects, and image contribution can be audited with explicit denominators. Perfect empty-passage behavior alongside incomplete supported answers is a reason to keep useful-answer coverage in the next evaluation. Refusing everything would satisfy the empty cases while failing the service objective.

4 Future Work

The next study will evaluate source applicability in native-language everyday information assistance. Swahili is the core target, with a gated Asante-compatible Twi replication and English reference questions. This recruitment plan differs from the completed Yoruba/Swahili diagnostic: no Twi experiment or new human assessment has been performed. AfriQA includes both Swahili and Asante Twi materials [5], but existing benchmark labels cannot replace language qualification or native-answer judgments.

On November 1, 2026, Prolific's public audience checker reported 701 participants active within 90 days with *First Language* set to Swahili, 1,872 with *Fluent languages* set to Swahili, and 70 with *First Language* set to Twi, with no other filter [8]. The Swahili pools may overlap and cannot be added. No direct Yoruba response option was found, which is not evidence of zero Yoruba speakers. Saved UI and selection records accompany the artifact. These mutable self-reported counts identify a possible recruitment route, not verified competence, reserved recruits or cultural representativeness [6, 9].

All qualification and human work will occur after funding, institutional ethics determination and consent. Paid screening will test reading, writing, source interpretation and relevant language variety; screened-out participants will also be compensated through the platform's permitted process [7]. We will seek 12 raters per language and require at least six qualified independent raters plus separate authorship/adjudication roles before freezing a language's test set. Authors will not rate their own scenarios. Twi work requires explicit compatibility with Asante materials. If qualification or costs fail, the scope will be reduced before locked outcomes: retain the Swahili core and English reference, and make Twi exploratory or omit it. Recruited contributors' residence and language history will be reported without treating diaspora participation as local community endorsement.

The planned collection has 120 source- and locality-grouped scenario families, 60 per retained language, partitioned into 40 development and 80 locked families. Domains include transport, educational opportunities, cultural events and routine administration; high-stakes advice is excluded. Four paired variants cover applicable evidence, withdrawn evidence, fluent inapplicable evidence, and approved spelling or code-switching changes. Inapplicable content may have apparently matching locality/date metadata but changed eligibility conditions. Independent reviewers must agree on the intended source status before freezing.

The primary comparison will use the same generation model, source snapshots, retrieval fusion and development-only thresholds. A strong reference already filters ordinary locality/date metadata. Additional qualification must justify proposition-scope and conflicting-content checks beyond that reference, rather than gain only against a metadata-blind baseline. Four configurations and two policies give $120 \times 4 \times 4 \times 2 = 3{,}840$ development-plus-locked responses, including 2,560 locked responses. One local and one hosted configuration will be designated for human audit by compatibility and cost before test outcomes. Current and nonempty inapplicable evidence across 80 locked families, two configurations and two policies yield 640 responses, independently double-rated for 1,280 ratings. Shared source/family clusters and

repeated raters will be accounted for in uncertainty estimates. Adequate answers, unsupported propositions, clarification and useful-answer coverage will remain distinct outcomes.

The funded development period will reconcile contributor roles, task duration and costs before the locked set. Months 1–3 establish qualification, permissions and rubric calibration; months 4–6 build and freeze the scenarios and interface; months 7–9 execute the paired comparison and human audit; months 10–12 complete adjudication and independent replay. A prospective target is a ten-percentage-point reduction in unsupported responses with no more than five points lost in adequate-answer coverage, with intervals and null findings retained. This is a research target, not an effect demonstrated by the present pilot. Permission-compatible scenarios, source identifiers, rubrics, model outputs and cost records would support a later expansion to additional qualified language panels.

5 Conclusion

Two small external-label studies separate response format, source extraction, question language, and cultural image use. All empty-passage instructions are followed, yet supported answer accuracy remains incomplete and question-language effects vary by model and item set. Images improve Yoruba cultural accuracy by three and four items in the paired comparisons, with individual losses retained. The evidence supports an auditable starting point for the proposed native-answer study, while leaving evaluator validity, source applicability and local-evidence service performance to future work.

Artifact preparation. Saurav K. Aryal directed the preliminary experiments with Codex assistance for implementation, execution, analysis, and manuscript preparation. Upstream datasets retain their creators' attribution and licenses; the public artifact provides source identifiers and download instructions rather than mirroring the image corpus or complete input passages.

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